Fall 2019 VE496
Lecture notes
Week 5

Problem statement

+

Purpose statement

=

Abstract
or
Executive summary

- To give readers a preview of what your report is about.

Key Words:

Abstract
Précis
Synopsis
Executive summary

- An **abstract** gives a short overview of a longer document
- *A* **précis** is a summary of essential points
- A **synopsis** of a movie summarizes the storyline without giving too many details.
- An **executive summary** is a report in miniature

Who writes abstracts?

- Researchers in all academic disciplines, including the sciences, the humanities, medical & legal studies

How short is an abstract?

- Anywhere from 300 to 500 words in length.

Table of Contents

ABSTRACT 2

ACKNOWLEDGEMENTS 3

I. INTRODUCTION 5

II. PROJECT MANAGEMENT 7

III. SYSTEM DESIGN AND ASSEMBLY 9

IV. MEASUREMENT RESULTS AND DISCUSSION 10

V. CONCLUSIONS 11

VI. REFERENCES 12

VII. APPENDIX 13

Examples:

International Journal of Educational Research 43 (2005) 432–445

International Journal of Educational Research

www.elsevier.com/locate/ijedures

Hybrid literacy texts and practices in technology-intensive environments

Paige D. Ware[a,*], Mark Warschauer[b]

[a]Southern Methodist University, 3108 Fondren, Box 750381, Dallas, TX 75275, USA

[b]Department of Education and Department of Informatics, 2001 Berkeley Place, Irvine, CA 92697-5500, University of California, Irvine, Irvine, CA 92697-5500, USA

Abstract

US youths' lives are increasingly divided between the academic requirements of school and immersion in new media and culture outside school. Educators can help bridge in-school and out-of-school literacy practices by encouraging students to engage with hybrid texts that draw on multiple modes of representation. In this paper, we analyze the 'disconnect' between academic literacy and new media, discuss the concept of hybridity as a way to bridge it, and provide a linguistically grounded analysis of students' hybrids texts and practices in two technology-intensive learning environments: a digital storytelling project in an after-school university-community collaborative and a one-to-one laptop program in an urban school district.

© 2006 Elsevier Ltd. All rights reserved.

1. Introduction

US youths' lives are increasingly divided between the academic requirements of school and immersion in new media and culture outside school. Educators encouraged by the potential for new media face the challenge of helping their students understand the standard ways of making meaning that are most influential in academia, while simultaneously engaging students in new forms of digital literacy. This challenge gives rise to the possibilities of hybridity, in which students cross traditional academic boundaries of genre, media, and modality in the production of texts. We focus in this

*Corresponding author. Tel.: +1 214 768 8623; fax: +1 214 768 8700.
E-mail address: pware@smu.edu (P.D. Ware).

0883-0355/$ - see front matter © 2006 Elsevier Ltd. All rights reserved.
doi:10.1016/j.ijer.2006.07.008

Journal of Power Sources 196 (2011) 10359–10368

Contents lists available at ScienceDirect

Journal of Power Sources

journal homepage: www.elsevier.com/locate/jpowsour

Design optimization of electric vehicle battery cooling plates for thermal performance

Anthony Jarrett, Il Yong Kim*

Queen's University, Kingston, Ontario, K7L 3N6, Canada

ARTICLE INFO

Article history:
Received 11 May 2011
Received in revised form 23 June 2011
Accepted 24 June 2011
Available online 2 July 2011

Keywords:
Electric vehicle battery
Cooling plate
Coolant flow channel
Thermal management
Temperature uniformity
Design optimization

ABSTRACT

The performance of high-energy battery cells utilized in electric vehicles (EVs) is greatly improved by adequate temperature control. An efficient thermal management system is also desirable to avoid diverting excessive power from the primary vehicle functions. In a battery cell stack, cooling can be provided by including cooling plates: thin metal fabrications which include one or more internal channels through which a coolant is pumped. Heat is conducted from the battery cells into the cooling plate, and transported away by the coolant. The operating characteristics of the cooling plate are determined in part by the geometry of the channel; its route, width, length, etc. In this study, a serpentine-channel cooling plate is modeled parametrically and its characteristics assessed using computational fluid dynamics (CFD). Objective functions of pressure drop, average temperature, and temperature uniformity are defined and numerical optimization is carried out by allowing the channel width and position to vary. The optimization results indicate that a single design can satisfy both pressure and average temperature objectives, but at the expense of temperature uniformity.

Crown Copyright © 2011 Published by Elsevier B.V. All rights reserved.

1. Introduction

To meet the expectations of the consumer, electric vehicles (EVs) must have a performance—especially considering range and battery reliability—comparable to that of a modern combustion-engine vehicle [1,2]. One of the current limitations is in the thermal control of the battery: operation at low temperature reduces the power output due to suppressed electro-chemical reactions, while elevated temperature accelerates corrosion leading to reduced battery life [3]. It is also important that the temperature within a battery cell is uniform: variations will cause the electrochemical reactions to proceed at different rates in different regions of the cell, thereby leading to incomplete energy utilization, and inefficient management of the battery life [3]. Any power consumed for the regulation of temperature reduces the power available to the primary vehicle functions, and so the efficient operation of a thermal management system is also desirable [4].

There are some situations (e.g. starting in cold environments) for which battery heating may be required [5], but for prolonged operation cooling will certainly be required. For batteries of low energy density, it may be possible to make use of air or passive cooling, but for higher energy density batteries, an active liquid cooled system gives the most effective and efficient thermal management [6].

In order to provide enough energy, an EV battery can be expected to occupy a large volume. For the most efficient use of space, a stack of rectangular laminate battery cells is often preferred, with temperature control by cooling plates inserted between alternate cells. These are typically thin metal pressings with one or more internal channels through which a coolant is pumped. The coolant absorbs heat from the cells and carries it away to a heat exchanger external to the battery. The performance of the cooling plate (in terms of heat transfer, temperature uniformity, power consumption, etc.) is determined by the fluid–solid interactions, and by the convective heat transfer from the plate into the coolant.

Literature on the specific subject of battery cooling plates is sparse, but several studies have considered EV battery cooling in general terms, or with other battery configurations. Ghosh et al. [7] studied air-cooling of cylindrical cells using CFD, and by modifying the airflow improved the battery temperature distribution. Likewise, Kim and Pesaran [6] studied the cooling of cylindrical cells with either air, oil, or a water–ethylene glycol mix. In an indirect liquid-cooled system water–ethylene glycol is superior to oil. However, mineral oil can be used in a direct-contact cooling system in which the heat transfer coefficients are much higher.

In the thermal control of fuel cells, the objectives of mean temperature and temperature uniformity are similarly important, and thermal control can also be addressed using cooling plates. Chen

* Corresponding author. Tel.: +1 613 533 3077; fax: +1 613 533 6489.
E-mail addresses: jarrett@me.queensu.ca (A. Jarrett), iykim@me.queensu.ca (I.Y. Kim).

0378-7753/$ – see front matter. Crown Copyright © 2011 Published by Elsevier B.V. All rights reserved.
doi:10.1016/j.jpowsour.2011.06.090

Abstract
Introduction

- To **orient** readers as to where your report is headed.

All images taken from http://cn.bing.com/images/search

Abstract

- **one paragraph long** (not an introduction)
- **complete in itself** (no reference numbers)

It should state in **summary** form,

- the general engineering background of your project
- your project objectives
- your design methodology
- measurement techniques
- observed facts & key understandings
- your conclusions

2 kinds of abstracts:

1. Descriptive abstract
2. Informative abstract

Descriptive abstract:

Informative abstract:

Abstracts are often reprinted in journals separated from the original papers, so they need to be *more or less* self-explanatory.

Descriptive abstract:

- Briefly states the main topic
- Makes general statements about project objectives
- Provides **NO** results or conclusions
- Often used by online journals to persuade readers to order a copy of the article

Informative abstract:

- Offers some details about method & materials used, results, plus conclusions
- Uses specialized language (jargon, nomenclature)
- Can stand alone as a substitute for the longer document

An informative abstract must pass the **Stand-alone Test**

An informative abstract must answer 4 questions to pass the "stand-alone" test:

1. What is the motivation behind the research?
→ ***Research purpose, general engineering background***

2. How did researchers carry out their study?
→ ***Procedure, methodology***

3. What did they find out?
→ ***Results***

4. What do their results mean?
→ ***Conclusions***

Key writing features in an abstract:

Precise

Concise

Succinct

- ✓ to express much in few words
- ✓ to be brief and to the point
- ✓ to eliminate unnecessary words

Some Examples

When to use passive voice

Active vs. passive voice

Active voice:

The workers fixed the leaking pipe under the street.

Passive voice:

The leaking pipes were fixed more than a week ago

To emphasize the action rather than the actor

- The proposal was finally approved by the planning committee.

To keep the subject and focus consistent throughout a passage

- The data processing department recently presented what proved to be a controversial proposal to expand its staff. After long debate, the proposal was approved by . . . .

To be tactful by not naming the actor

- The procedures were somehow misinterpreted.

To describe a condition in which the actor is unknown or unimportant

- Every year, thousands of people are diagnosed as having cancer.

To create an authoritative tone

- Visitors are not allowed after 9:00 p.m.

Active vs. passive voice

1. A team of 5 workers fixed the leaking pipe under the street.
2. The leaking pipes were fixed more than a week ago.

3. The manager held a meeting to announce the new polices.
4. The new polices were formally announced on Oct. 12.

5. The graduate program admits only 10 students a year.
6. Only 10 students are admitted into the graduate program a year

Reporting writing styles:

Formal reports have a higher degree of polish and production quality.

Informal reports are often distributed in a memo format as an email attachment.

Impersonal vs. personal writing

- Objectivity
- Believability

Personal

Having studied the advantages and disadvantages of using coupons, I recommend that your company not adopt this practice. If you used coupons, you would have to absorb their cost. You would also have to hire additional employees to take care of the increase in sales volume.

Impersonal

A study of the advantages and disadvantages of using coupons supports the conclusion that the Mills Company should not adopt this practice. The coupons themselves would cost extra money. Also, use of coupons would require additional personnel to take care of the increase in sales volume.

Mixing active and passive voice

An impingement jet ***occurs*** when a high coolant mass ***is ejected*** from a hole impinging on a high heat transfer surface.

The fluid in the center ***pushes*** neighboring fluid, leading to the curve as shown in Figure 1.1. Due to this pushing of fluid, an increase in turbulence is present. Vortices ***are also formed*****.**

Verb tenses

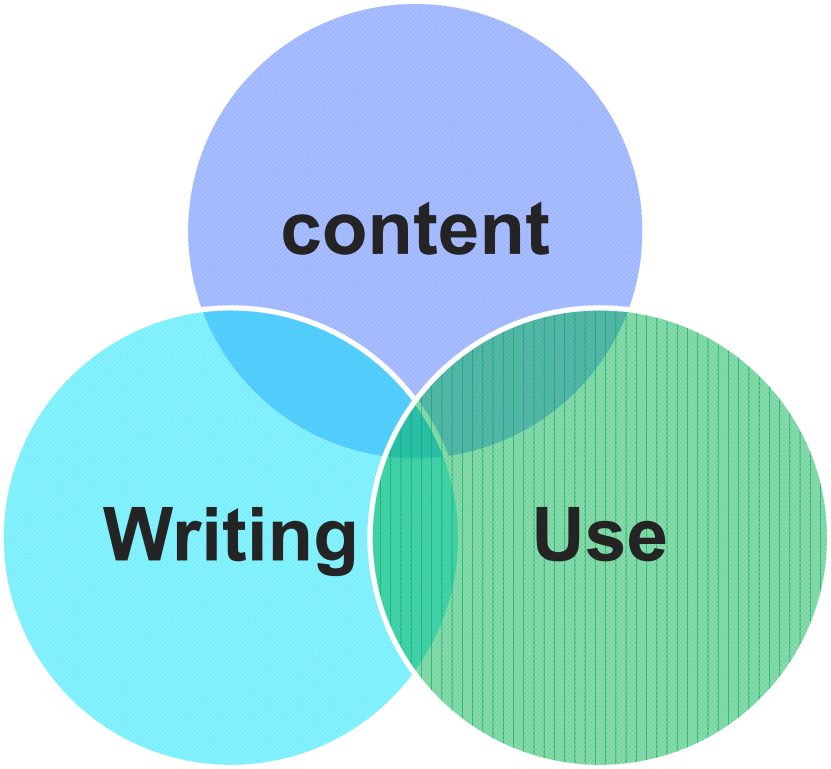
- **Simple present:**

*Heat **is** conducted from the battery cells into the cooling plate.*

- **Present perfect:**

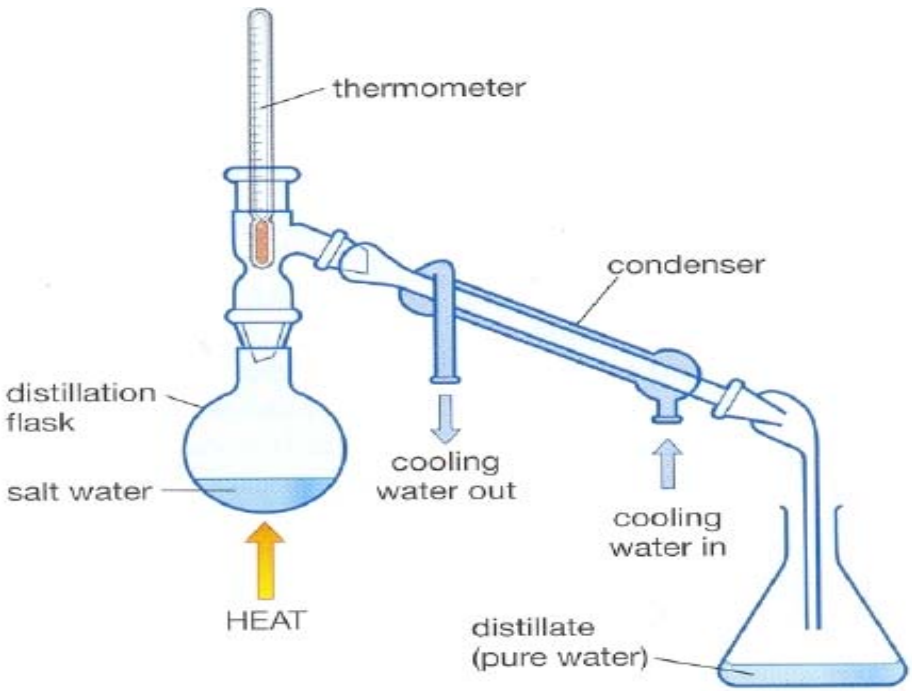
*The analyses **have been used** to predict the velocity flow field for a three-wick candle contained within a ceramic vessel.*

Content and writing CANNOT be separated



Main Benefit of writing

"The act of writing forces us to *distill* vague notions into clear ideas."



Thank you!

www.umji.sjtu.edu.cn